Replicating Table 1, Panel B from Chernozhukov et al. (2017)

Aaron Chotzen-Jenner

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1 The Original Paper

Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, and Newey (2017), "Double/Debiased/Neyman Machine Learning of Treatment Effects," *American Economic Review: Papers & Proceedings*, 107(5): 261–265, introduces the Double/Debiased Machine Learning (Double ML) estimator for causal inference in settings where high-dimensional nuisance parameters must be estimated. The central research question is how to obtain valid, root-N consistent estimates of a treatment effect parameter θ_0 when the confounding functions are estimated using flexible machine learning methods that introduce regularization bias. In simpler terms, the paper explains how to measure the true effect of a treatment when machine learning is used to control for other factors that may influence the outcome.

The authors approach this problem with two main methods. First, the estimating equation is made Neyman-orthogonal, meaning it is locally insensitive to small errors in the nuisance parameter estimates. In other words, the equation is designed in a way that small mistakes in the machine learning predictions have very little impact on the final treatment effect estimate. Second, cross-fitting is used to split the sample into folds, training the nuisance functions (everything else that needs to be estimated besides the main effect) on held-out data to avoid overfitting bias. Together, these two steps remove the regularization bias and therefore not distorted by error from the machine learning models.

2 The Result of Interest

The specific result I replicate is the Mean Average Treatment Effect (ATE) reported in Table 1, Panel B, row "ATE (2 fold)," Ensemble column of the online appendix. This result corresponds to the partially linear model with 2-fold cross-fitting. The paper reports a Mean ATE of \$9,043 with a standard error of 1,432.

This result is the headline finding of the paper: after flexibly controlling for income and other confounding factors, 401(k) eligibility increases net financial assets by roughly \$9,000. The underlying model estimates the relationship between financial assets, 401(k) eligibility, and a set of demographic and economic control variables. Rather than relying on a single machine learning algorithm, the Ensemble method combines predictions from several approaches, including Lasso, Random Forests, Boosting, and Neural Networks. It assigns more weight to the methods that perform best out-of-sample, as the goal is to obtain a more reliable estimate than any single method alone.

I chose this result to replicate because it is the paper's primary empirical finding, corresponds to the default specification in the replication code (`Master.R`), and illustrates the central idea of Double ML, which is using machine learning to control for complex confounding while still producing a credible estimate of a causal treatment effect.

3 Data and Methods

The data come from the 1991 Survey of Income and Program Participation (SIPP), the same dataset used in Chernozhukov and Hansen (2004). The sample contains 9,915 observations. The outcome variable `net_tfa` is net total financial assets, measured in dollars. The treatment variable `e401` is a binary indicator for eligibility to enroll in a 401(k) plan. The nine covariates used in the model are age, income, years of education, family size, and binary indicators for married status, two-earner household, defined benefit pension, IRA participation, and home ownership.

`preprocess.R` loads the raw Stata file `sipp1991.dta` from `input/`, selects the eleven relevant variables, and writes a clean CSV to `temp/clean_data.csv`. `analysis.R` loads that file, sources the original helper functions `ML_Functions.R` and `Moment_Functions.R` from `input/`, and calls `DoubleML()` with the partially linear model specification and 2-fold cross-fitting. To reduce computation time, the estimation is repeated over 10 random sample splits rather than the 100 used in the original paper. I used parallel computation across 4 cores of my computer to accelerate estimation. The Mean ATE, Median ATE, and their standard errors are computed following the paper's formulas exactly. Results are written to `output/tables/main_result.tex`, which is inputted directly into this document.

4 Replication Results

Table 1: Replication of Table 1, Panel B (Partially Linear Model, 2-fold) from Chernozhukov et al. (2017)

	This Replication	Paper (Ensemble)
Mean ATE ($)	8970.03	9,043
Std. Error	1406.97	1,432
95% CI	[6212.38, 11727.68]	—
Median ATE ($)	9017.29	9,061
Median SE	1406.97	1,343

Note: Based on 10 sample splits, 2-fold cross-fitting.

Paper values from Table 1, Panel B, Ensemble column.

Table 2: Replication vs. Paper: Table 1, Panel B (Partially Linear Model, 2-fold)

	RLasso	Reg. Tree	Random Forest	Boosting	Neural Net	Ensemble
This Replication						
Mean ATE ($)	7603	8583	8757	8770	8792	8970
	(1729)	(1518)	(1426)	(1379)	(1434)	(1407)
Chernozhukov et al. (2017)						
Mean ATE ($)	7718	8745	9180	8768	9040	9043
	(1796)	(1488)	(1526)	(1451)	(1494)	(1432)

Note: Standard errors in parentheses. Replication based on 10 sample splits, 2-fold cross-fitting. Paper values from Table 1, Panel B of the online appendix.

The replication estimate reported in Table 1 closely matches the corresponding result in Chernozhukov et al. (2017). The estimated average treatment effect of 401(k) eligibility on net financial assets is very similar to the published estimate, and the associated standard errors are nearly identical. Small differences are expected because the Double ML procedure relies on repeated random sample splitting and machine learning methods whose predictions vary slightly across replications. The original study averages results over 100 random sample splits, while this replication uses only 10 splits. This was done to keep computation time manageable on a personal computer. With fewer splits, individual random partitions of the data have a larger influence on the final estimate, leading to slightly more variation across runs. Despite these differences, the replication estimate remains within one percent of the published result, suggesting that the implementation successfully reproduces the paper's central empirical finding that 401(k) eligibility substantially increases net financial assets.

Table 2 provides a comparison of the individual machine learning methods used in the replication and the corresponding estimates reported in the original paper. While there are modest differences in the point estimates across methods, the overall pattern is very similar. In both the replication and the paper, RLasso produces the lowest estimated treatment effect, Random Forest and Neural Net produce some of the highest estimates, and the Ensemble method gives results near the upper end of the range. The standard errors are also broadly comparable across all methods. These similarities suggest that the replication successfully reproduces not only the headline Ensemble estimate but also the relative performance of the underlying machine learning approaches. The consistency of results across methods reinforces one of the paper's central conclusions: when combined with orthogonalization and cross-fitting, different machine learning algorithms tend to produce similar estimates of the treatment effect.

5 What I Learned

Once the helper functions were sourced correctly, reproducing the estimation was fairly straightforward. The hardest part was path management, as the original code assumed all files lived in the same working directory, while the Gentzkow-Shapiro structure separates `input/`, `code/`, and `temp/`. In particular, `Moment_Functions.R` internally sources `ML_Functions.R` with a bare filename, which required patching to use an absolute path. This was a pure reproducibility issue: code that works perfectly in the author's environment can break when placed in a different directory structure. This happened to me before I made the fix. If I were the original authors, the one thing that would have made replication easier is absolute rather than relative paths in the helper files, along with a brief README note specifying how the scripts should be run.

More broadly, this replication taught me that Double ML is amazingly stable. Six very different machine learning methods (RLasso, Trees, Random Forest, Boosting, Neural Net, and Ensemble) all produce estimates between approximately \$7,900 and \$9,200, with substantial overlap in their confidence intervals. This stability is exactly what the Neyman orthogonality theory predicts, and seeing it empirically with real-world data was something I could have learned only from doing this replication.

References

[1] Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., and Newey, W. (2017). "Double/Debiased/Neyman Machine Learning of Treatment Effects." *American Economic Review: Papers & Proceedings*, 107(5): 261–265.